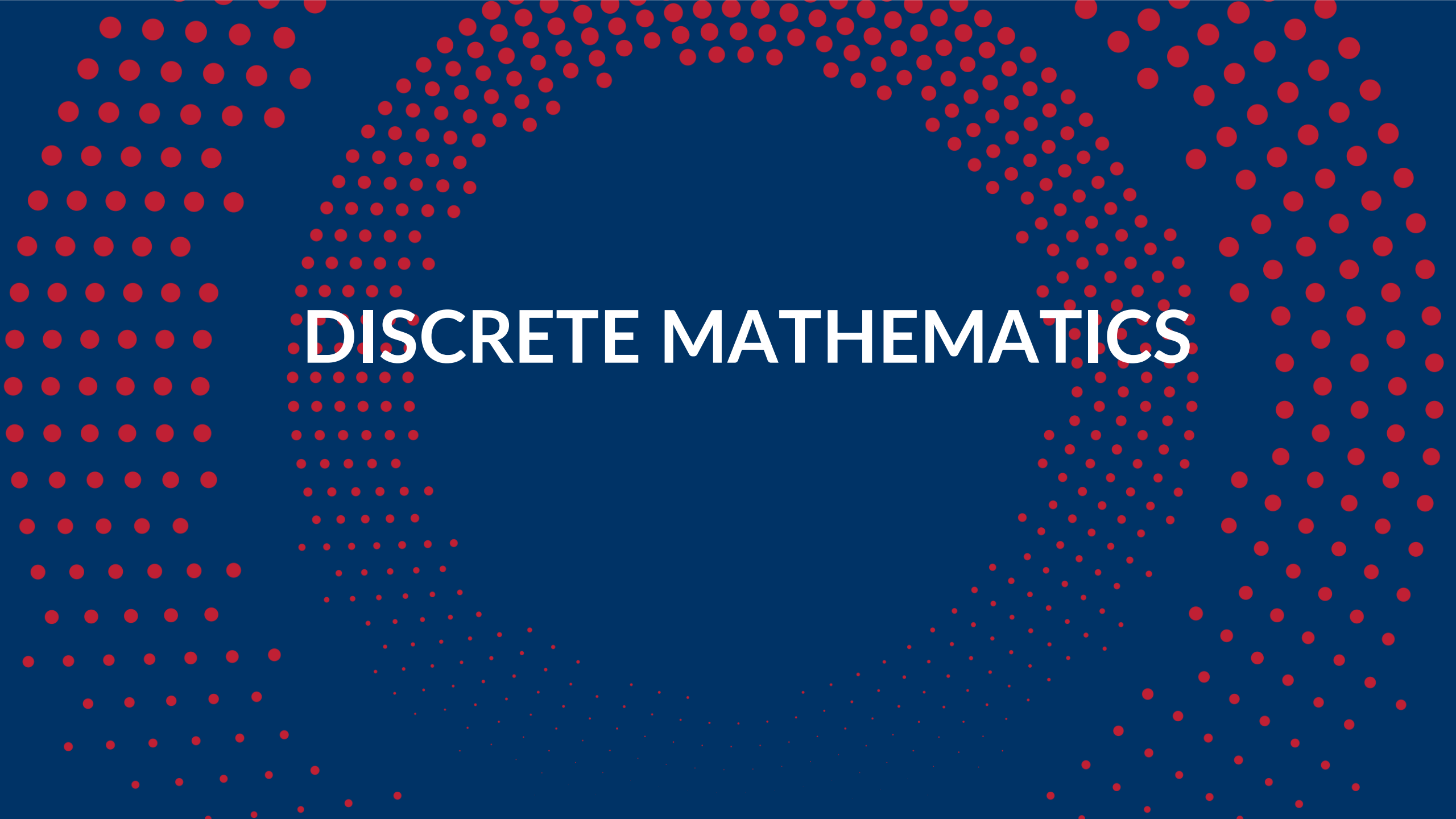




HUST

ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

ONE LOVE. ONE FUTURE.



DISCRETE MATHEMATICS



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OF SCIENCE AND TECHNOLOGY

DISCRETE MATHEMATICS

ONE LOVE. ONE FUTURE.

PART1: COMBINATORIAL THEORY

PART 2: GRAPH THEORY

Contents of part 1

- Chapter 0: Sets, Relations, Functions
- Chapter 1: Counting problem
- Chapter 2: Existence problem
- Chapter 3: Enumeration problem
- Chapter 4: Combinatorial optimization problem

Chapter 0: Sets, Relations, Functions

- Sets
- Relations
- Functions



- Definitions
- Venn diagram
- Set operations
- Equality on sets
- Power Set
- Partition

- **Definitions**
- Venn diagram
- Set operations
- Equality on sets
- Power Set
- Partition

Definitions

- **Definition:**
 - A set is an unordered collection of (unique) objects
 - The objects in a set are called elements or members of a set.A set is said to contain its elements.
- Notation, for a set A:
 - $x \in A$: x is an element of A
 - $x \notin A$: x is not an element of A

Example:

$V = \{a, e, i, o, u\}$ (vowels in English)

C = all students subscribed to IT3020E in Winter 2020

Note:

- We often denote sets with capitals
- Brackets are used to define the set. $\{.\}$

Our first concern will be how to describe a set; that is, how do we most conveniently describe a set and the elements that are in it? Sets can be defined in various ways:

1. Set extension
2. Set intension

Definitions

- A set is defined in **extension** when you enumerate all the elements:

$$O = \{0, 2, 4, 6, 8\}$$

- The **set-builder** notation

$$A = \{x \mid \text{conditions}(x)\}.$$

this could be read as “all x such that the conditions hold true”.

Example: $O = \{x \mid (x \in \mathbb{Z}) \wedge (x = 2k) \text{ for some } k \in \mathbb{Z}\}$

reads: O is the set that contains all x such that x is an integer and x is even

- A set is defined in **intension** when you give its set-builder notation

$$O = \{x \mid (x \in \mathbb{Z}) \wedge (0 \leq x \leq 8) \wedge (x = 2k) \text{ for some } k \in \mathbb{Z}\}$$

Definitions

- There is a set with no elements. It is called the **empty set** (or *null set*) and denoted $\{\}$ or $\emptyset$.
- A set that has one element is called a **singleton set**.
 - For example: $\{a\}$, with brackets, is a singleton set
 - a , without brackets, is an element of the set $\{a\}$
- Note the subtlety in $\emptyset \neq \{\emptyset\}$??why
 - The left-hand side is the empty set
 - The right hand-side is a singleton set, and this set contains a set

Definitions

- If there are exactly n distinct elements in a set S , with n is a nonnegative integer, we say that:
 - S is a **finite set**, and
 - The **cardinality** of S is n . Notation: $|S| = n$.
- **Definition.** A set is a **finite set** if it has a finite number of elements. A set that is not finite is an *infinite set*.
- Let A be a finite set. The number of different elements in A is called its **cardinality** and is denoted by $|A|$. Other notations commonly used for the cardinality of A are $N(A)$, $\#A$.
If A be a infinite set, then we write $|A| = \infty$.

Example:

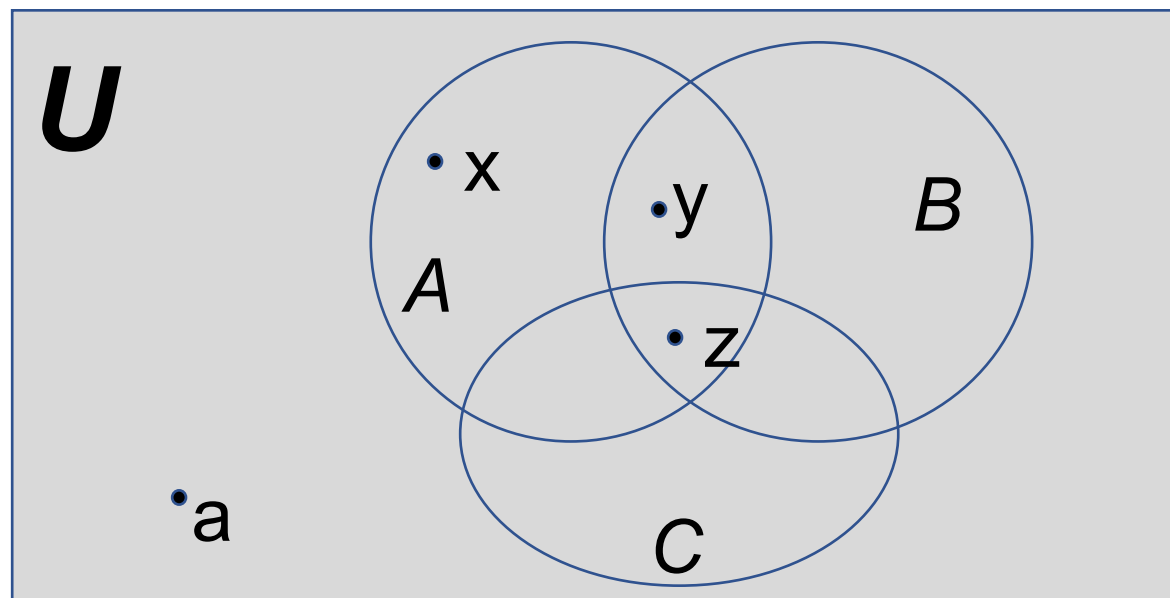
- $|\emptyset| = 0$ since $\emptyset$ contains no elements.
- $|\{\pi, 2, \text{Newton}\}| = 3$.
- If $N_n = \{0, 1, \dots, n\}$ then $|N_n| = n + 1$.
- $|\{n: n \text{ is a prime number}\}| = \infty$.

- Definitions
- **Venn diagram**
- Set operations
- Equality on sets
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- Partition

Venn diagram

- A set can be represented graphically using a **Venn Diagram**
- The universal set U is represented by the interior of a rectangle
- The sets by region inside the rectangle and elements which belong to a given set are placed inside the region representing it.
- If an element belongs to more than one set in the diagram, the two regions representing the sets concerned must overlap and the element is placed in the overlapping region.

In this way the picture represents the relationships between the sets concerned.



- Definitions
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Set operations

- Arithmetic operators ($+$, $-$, $\times$, $\div$) can be used on pairs of numbers to give us new numbers
- Similarly, set operators exist and act on two sets to give us new sets:
 1. Union
 2. Generalized union
 3. Intersection
 4. Generalized intersection
 5. Set difference
 6. Set complement

- Definitions
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Equality on sets

Equality	Name
$A \cup \emptyset = A$ $A \cap U = A$	Identity laws (Đồng nhất)
$A \cup U = U$ $A \cap \emptyset = \emptyset$	Domination laws (Trội)
$A \cup A = A$ $A \cap A = A$	Idempotent laws (Lũy đẳng)
$\overline{(\overline{A})} = A$	Complementation laws (Bù)

Equality on sets

Equality	Name
$A \cup B = B \cup A$ $A \cap B = B \cap A$	Commutative laws (Giao hoán)
$A \cup (B \cap C) = (A \cup B) \cap C$ $A \cap (B \cup C) = (A \cap B) \cup C$	Associative laws (Kết hợp)
$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	Distributive laws (Phân phối)
$\overline{A \cap B} = \overline{A} \cup \overline{B}$ $\overline{A \cup B} = \overline{A} \cap \overline{B}$	De Morgan's laws (Luật De Morgan)

Power set

- Power set (or powerset) of a set A , denoted by $P(A)$ (or 2^A), is the set of all subsets of A
- Example $A = \{1, 2, 3\}$, $P(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$
- Property: $|P(A)| = 2^{|A|}$
- Exercise: For all sets A and B , the statement “ $A \subseteq B$ if and only if $P(A) \subseteq P(B)$ ” is TRUE or FALSE?

Partition

- Let $X_1, X_2, \dots, X_m$ be subset of X . We say $X_1, X_2, \dots, X_m$ is a **partition** of X if:
 - $X = X_1 \cup X_2 \cup \dots \cup X_m$;
 - $X_i \cap X_j = \emptyset, i \neq j$.
 - $X_i \neq \emptyset, i=1, \dots, m$

Example: $X = \{1, 2, 3, 4\}$

$E = \{\{1, 2\}, \{3, 4\}\}$ is a partition of X

$F = \{\{1, 2, 3\}, \{3, 4\}\}$ is not a partition of X

Chapter 0. Set, Relations, Functions

1. Set

2. Relations

3. Functions



Relations

- **Ordered pair**
- Binary relation
- Relation representation
- Properties of binary relations

Ordered pairs

- An *ordered pair* is a pair of objects with an order associated with them.
- In general (x, y) is different from (y, x) .
- **Definition (equality of ordered pairs):** Two ordered pairs (a, b) and (c, d) are equal **if and only if** $a = c$ and $b = d$.
- Example: if the ordered pair (a, b) is equal to $(1, 2)$, then $a=1$, and $b=2$.

Binary Relations

- **Definition:** Let A and B be sets. A binary relation from a set A to a set B is a set of ordered pairs (a, b) where a is an element of A and b is an element of B .
 - A binary relation from A to B is a subset $R \subseteq A \times B$
 - A **relation on a set A** is a relation from A to A , i.e., a subset $R \subseteq A \times A$
- **Notation:** When an ordered pair (a, b) is in a relation R , we write $a R b$, or $(a, b) \in R$. It means that element a is related to element b in relation R . We will write $a \bar{R} b$ when a element a is not related to element b in relation R .

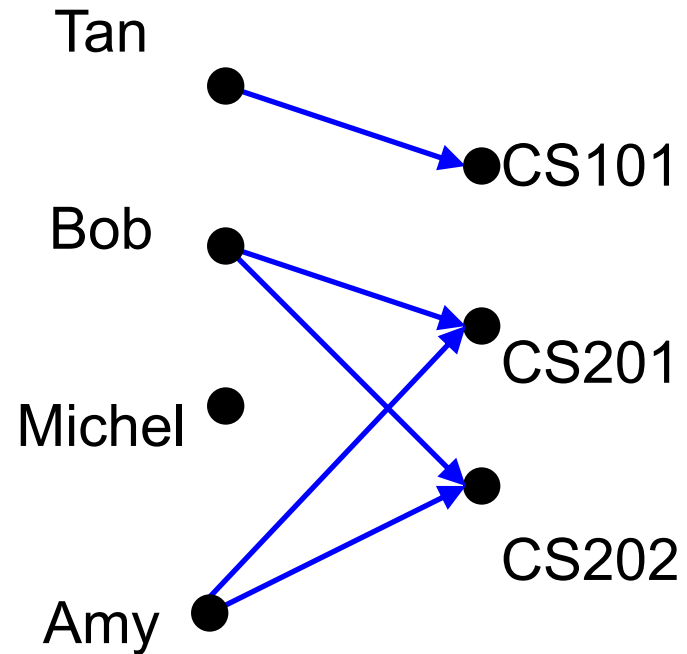
Binary Relations

Example:

- Let A be the students in the CS major
 - $A = \{\text{Tan, Bob, Michel, Amy}\}$
- Let B be the courses the department offers
 - $B = \{\text{CS101, CS201, CS202}\}$
- We specify relation $R \subseteq A \times B$ as the set of students of A enrolled in classes of B
 - $R = \{ (\text{Tan, CS101}), (\text{Bob, CS201}), (\text{Bob, CS202}), (\text{Amy, CS201}), (\text{Amy, CS202}) \}$

Relation representation

We can represent relations graphically:



We can represent relations in a table:

	CS101	CS201	CS202
Tan	X		
Bob		X	X
Michel			
Amy		X	X

Properties of Binary Relations

- A binary relation $R \subseteq A \times A$ is called
 - Reflexive iff $(x, x) \in R$
 - Symmetric iff $(x, y) \in R \rightarrow (y, x) \in R$
 - Antisymmetric iff $((x, y) \in R \wedge (y, x) \in R) \rightarrow x = y$
 - Transitive iff $(x, y) \in R$ and $(y, z) \in R \rightarrow (x, z) \in R$
- A binary relation is called equivalence relation iff it is reflexive, symmetric, and transitive
- Example
 - Equality on a set of numbers: $a = a$ (reflexive); $a = b \rightarrow b = a$ (symmetric); $a = b \wedge b = c \rightarrow a = c$ (transitive)
 - Congruence modulo on integers: $a \equiv b \pmod{n}$
 - “Same shape” on the set of triangles: “triangle A is similar to triangle B” is an equivalence relation

Properties of Binary Relations

- Exercise: We define a relation $\sim$ on the set $\mathbb{Z}$ of integers by: $x \sim y \equiv \gcd(x,y) = 1$. Statement " $\sim$ is an equivalence relation on $\mathbb{Z}$ " is TRUE or FALSE?

Chapter 0. Set, Relations, Functions

1. Set
2. Relations
- 3. Functions**



- Definition
- Function representations
- Some types of functions

- **Definition**
- Function representations
- Some types of functions

Definition

- **Definition:** A function f from a set A to a set B , denote it by $f: A \rightarrow B$, is a relation from A to B that satisfies:
 - for each element a in A , there is an element b in B such that (a, b) is in the relation, and
 - if (a, b) and (a, c) are in the relation, then $b = c$.

A function is also called a *mapping* or a *transformation*.

Definition

- The **image of the set S under function $f: A \rightarrow B$** , denoted by $f(S)$ is:

$$f(S) = \{ f(a) \mid a \in S \}$$

- The **image of the domain under function $f: A \rightarrow B$** , denoted by **range f** is:

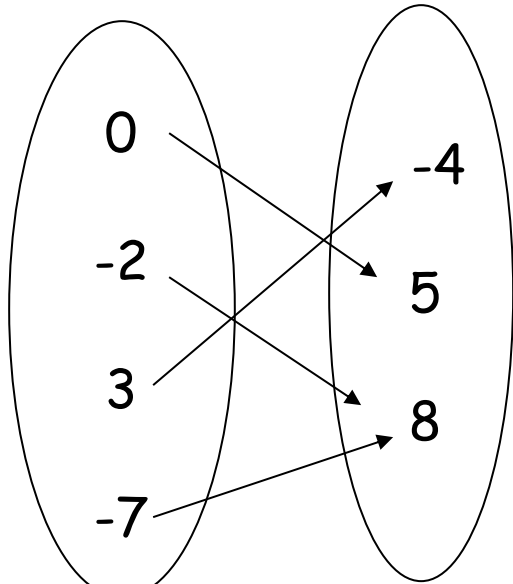
$$\text{range } f = f(A)$$

(is also called the **range of f**)

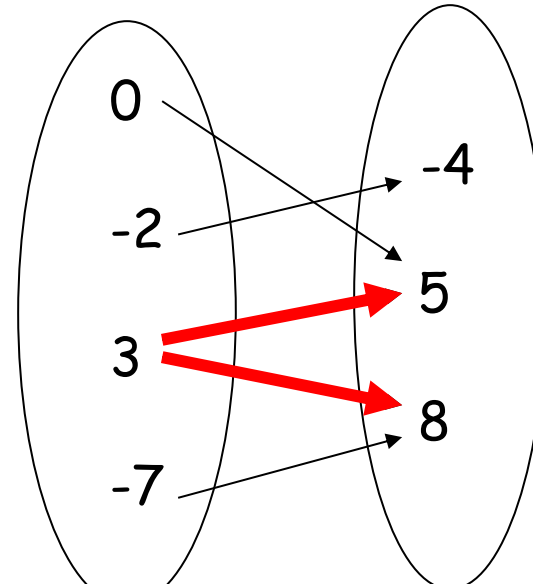
In general case: $\text{range } f = f(A) \subseteq B$.

Definition

Example: Are the following relations also functions?



Each element of the domain is only used once.



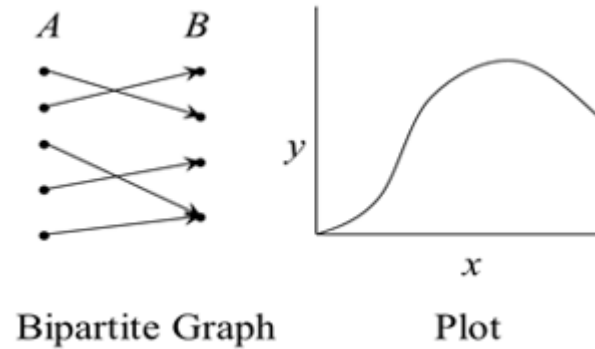
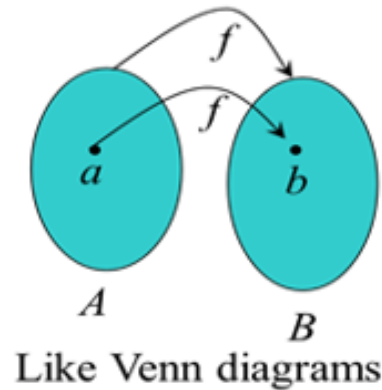
3 is used twice.

- Definition
- **Function representations**
- Some types of functions

Relation representations

- Functions can be represented four different ways:

- Mapping
- Graph
- Table
- Matrix



a	a_1	a_2	...	a_m
$f(a)$	$f(a_1)$	$f(a_2)$	...	$f(a_m)$

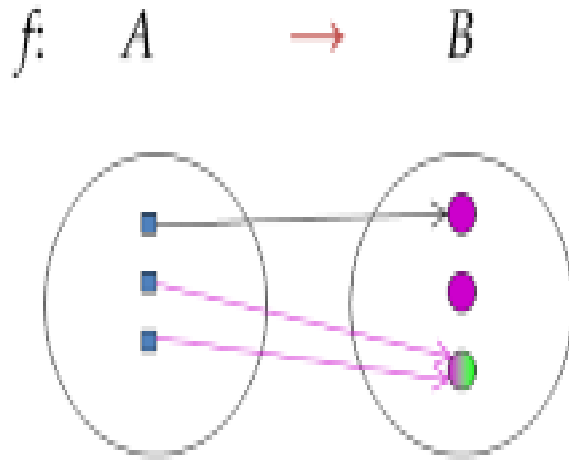
The function $f: A \rightarrow B$ can be defined by a matrix $A_f = \{a_{ij}\}$ size of $m \times n$ where

$$a_{ij} = \begin{cases} 1, & \text{if } b_j = f(a_i), i = 1, \dots, m; j = 1, \dots, n. \\ 0, & \text{otherwise} \end{cases}$$

- Definition
- Function representations
- **Some types of functions**

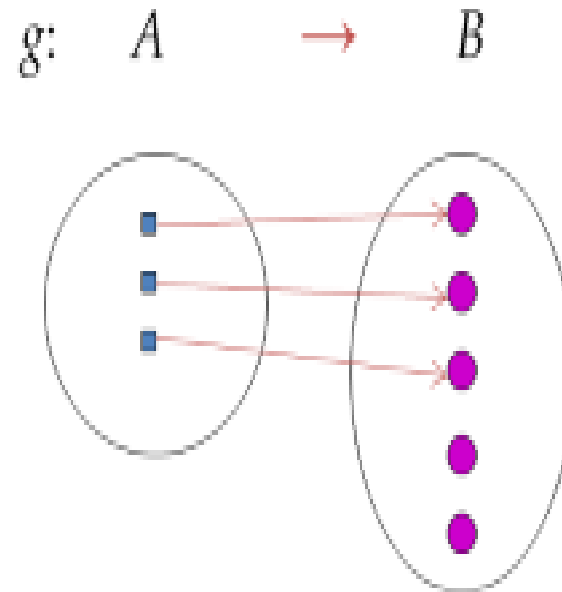
Some types of function

- **Injective:** A function f from a set A to a set B is said to be **injective (one-to-one)** if and only if:
for all elements $a_1, a_2 \in A$
if $f(a_1) = f(a_2)$ then $a_1 = a_2$



The function f is not injective

$$a_1, a_2 \in A, a_1 \neq a_2 \Rightarrow f(a_1) = f(a_2)$$



The function g is injective

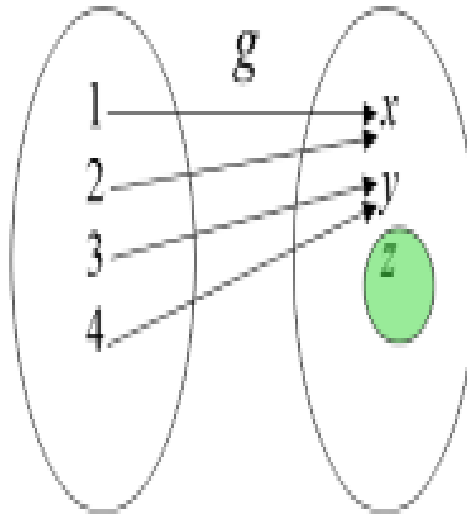
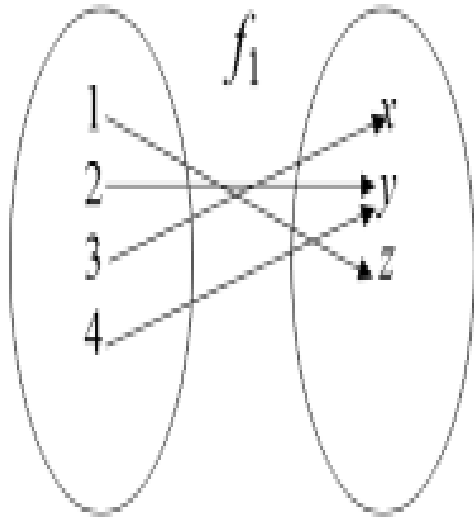
Some types of function

- **Surjective:** A function f from a set A to a set B is said to be **surjective (onto)**, if and only if: $\forall b \in B, \exists a \in A: b = f(a)$.

that is: f is onto if and only if $f(A) = B$.

Example: $A = \{1, 2, 3, 4\}$ and $B = \{x, y, z\}$. Then the functions:

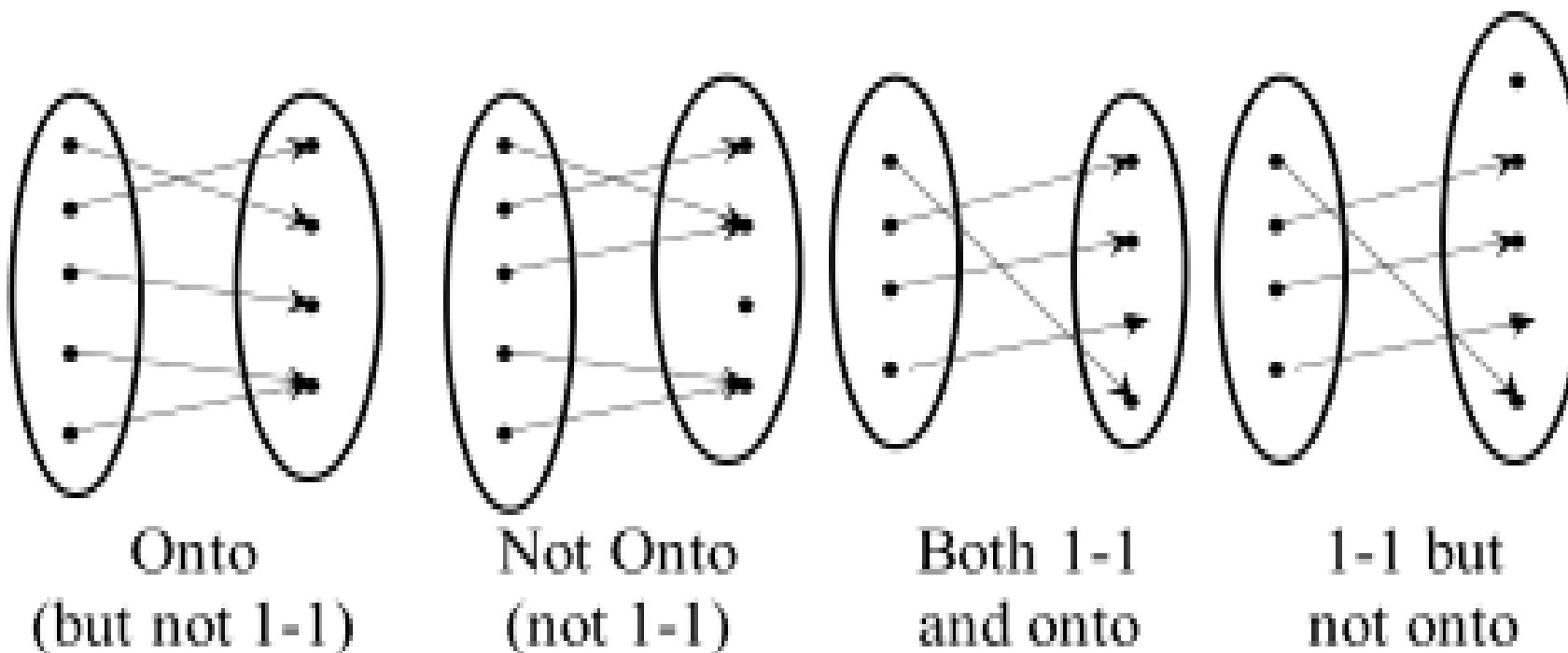
$$f_1 = \{(1, z), (2, y), (3, x), (4, y)\}; \quad g = \{(1, x), (2, x), (3, y), (4, y)\}$$



The function g is not onto because $g(A) = \{x, y\} \neq B$

Some types of function

- **Bijection:** A function is called a *bijection*, if it is injective (1-1) and surjective (onto).



Exercise

- $f: A \rightarrow B$ is a function from a set A to a set B , $S \subseteq A$, and $T \subseteq A$. Statement “ $f(S \cup T) = f(S) \cup f(T)$ ” is TRUE or FALSE ?

Exercise

- $f: A \rightarrow B$ is a function from a set A to a set B , $S \subseteq A$, and $T \subseteq A$. Statement “ $f(S \cap T) = f(S) \cap f(T)$ ” is TRUE or FALSE?

A graphic on the left side of the slide. It features a dark blue background with a large, stylized circular pattern made of red dots of varying sizes, creating a halftone or mesh effect. The word "HUST" is centered within this pattern in a white, bold, sans-serif font.

HUST

THANK YOU !